Part A: Literature Review and Public Dataset

CAP182 Capstone Project: STADIOEquities

Publication 1

Title	*A data-driven approach to predict the success of bank telemarketing*
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22–31. doi: 10.1016/j.dss.2014.03.001.
Link	https://doi.org/10.1016/j.dss.2014.03.001 Open-access post-print: https://repositorio.iscte-iul.pt/bitstream/10071/9499/5/dss_v3.pdf
Modelling	The study compared logistic regression, decision trees, neural networks and support vector machines (SVM). Model performance was evaluated using the area under the ROC curve (AUC) and cumulative LIFT curves. A rolling time-window evaluation approach was also used.
Preprocessing techniques	The study initially considered a large set of 150 candidate features covering client demographic information, campaign-contact information and macroeconomic variables. A semi-automatic feature selection step reduced these to a set of 22 relevant features. The study also used a rolling time-window approach to evaluate model performance across different periods.
Datasets used	Portuguese bank telemarketing campaign data. The dataset was subsequently made publicly available through the UCI Machine Learning Repository as the Bank Marketing dataset.
Key takeaways	1. Customer and campaign-related information can be useful for predicting financial-product conversion outcomes. 2. Time-aware evaluation can be important when working with data collected over different periods. This project will use a time-ordered split when evaluating its models. 3. Comparing several modelling techniques can help identify an appropriate approach rather than relying on a single model. This project will therefore compare two different models. **Relevance to STADIOEquities:** The study predicts whether a contacted customer converts to a financial product, which is a close analogue of predicting whether a newly registered customer will make a first deposit. It is also the source of the public dataset used in Part B. Most customers in that dataset had no earlier contact with the bank, so the prediction is largely made for customers about whom little history exists, as it will be for new registrants. The main difference is that the campaign information is richer than the information available at registration.

Publication 2

Title	*Predicting Purchase Decisions in Mobile Free-to-Play Games*
Harvard reference	Sifa, R., Hadiji, F., Runge, J., Drachen, A., Kersting, K. and Bauckhage, C. (2015) 'Predicting purchase decisions in mobile free-to-play games', *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, 11(1), pp. 79–85. doi: 10.1609/aiide.v11i1.12788.
Link	https://doi.org/10.1609/aiide.v11i1.12788 https://ojs.aaai.org/index.php/AIIDE/article/view/12788
Modelling	**Classification** (will the player ever buy?): decision trees (reduced-error pruning, Gini index), random forests (151 trees, square root of the number of features tried per split, Gini index) and support vector machines (polynomial and RBF kernels, tuned by grid search). **Regression** (how many purchases?): Poisson regression trees compared with a Poisson regression baseline. Evaluation used ten-fold cross-validation with recall, precision, G-mean, F-score and RMSE. Random forests trained on oversampled data performed best, while decision trees were used to produce rules a business can act on.
Preprocessing techniques	Data from several tables was merged and observation windows of 1, 3 and 7 days after installation were built, so that only early behaviour was used. Telemetry features (logins, game rounds, playtime, interactions, spend) were combined with engineered features such as the time between sessions and logins, and the mean, deviation and correlation of behaviour over time. The country variable (almost 200 categories) was collapsed into country segments. Fewer than 2% of players were purchasers, so SMOTE-NC, a version of SMOTE that handles mixed numeric and nominal attributes, was used to oversample the minority class.
Datasets used	A proprietary dataset from a mobile puzzle game published by Wooga. It was a random sample of one week of new installs (over 100,000 players) followed for 30 days, and the classification experiments used a 10,000-player sample. The data is not publicly available.
Key takeaways	1. Purchases can be predicted for brand-new users from a short early window, but performance depends strongly on how much early behaviour is available. The F-score of the best classifier rose from 0.188 (1 day) to 0.384 (3 days) and 0.522 (7 days), because most players look identical at the start. This project should therefore expect modest performance from first-contact information, and should compare against a baseline. 2. A very small purchasing class (under 2%) required explicit imbalance handling. This project will handle its 11% positive class with class weighting rather than SMOTE-NC, because class weighting does not create synthetic customers from mixed categorical and numeric attributes. 3. An interpretable decision tree gave rules that a business can act on, while a random forest gave the best predictions. This supports comparing an interpretable model with a tree-based model, and judging whether the more complex model is worth its cost. **Relevance to STADIOEquities:** This is the closest match to the client problem. It predicts a first monetary action (a purchase) for users who have just joined and have little or no history, and it uses early behaviour, as the first-deposit prediction will. The differences are that the data is game telemetry rather than banking data, the purchasing rate is much lower, and the dataset is proprietary.

Publication 3

Title	*Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks*
Harvard reference	Sakar, C.O., Polat, S.O., Katircioglu, M. and Kastro, Y. (2019) 'Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks', *Neural Computing and Applications*, 31, pp. 6893–6908. doi: 10.1007/s00521-018-3523-0.
Link	https://doi.org/10.1007/s00521-018-3523-0
Modelling	Two modules were used. The first predicts whether a visitor will buy, using page-view data from the visit together with session and visitor information. It compared random forest, support vector machine and multilayer perceptron (MLP, trained with resilient backpropagation) classifiers. The MLP gave the best accuracy and F1 score. The second module used a long short-term memory (LSTM) recurrent network on the sequence of clicks to estimate the probability that a visitor leaves without buying within a set time. Combining both modules identifies visitors who intend to buy but may leave.
Preprocessing techniques	Oversampling was applied to address the imbalance between buying and non-buying sessions, and feature selection was used to improve classifier performance. The abstract reports that combining click-stream features with session-based features improved the success rate. Features were built per session from page-view counts and durations, and from Google Analytics measures such as bounce rate, exit rate and page value, together with visitor and session attributes (the public dataset is described below).
Datasets used	The Online Shoppers Purchasing Intention Dataset (Sakar and Kastro, 2018): 12,330 sessions with 17 features and a binary target (Revenue), of which 15.5% are purchases. Each session belongs to a different user over one year, so that the data is not biased towards a particular user profile, campaign, special day or period. It is public on the UCI Machine Learning Repository under a CC BY 4.0 licence (doi: 10.24432/C5F88Q).
Key takeaways	1. Because every record is a different user, there is no personal history to draw on, and the model must work from early, observable behaviour and context. This reinforces the decision in this project to exclude *duration* from the Bank Marketing data, since it is only known after the contact has taken place. 2. Oversampling and feature selection were useful on a target with 15.5% positives. This project will check the class balance, handle the imbalance, and report F1 and ROC-AUC as well as accuracy. 3. Combining behavioural and session-level information worked better than either alone. This supports the session and timing fields requested in the SS1 data request, which the public Bank Marketing data cannot supply. **Relevance to STADIOEquities:** The paper predicts a purchase from the early behaviour of visitors who have no history with the site, which mirrors predicting a first deposit from the early behaviour of new registrants. Its dataset is public and built on one-session-per-user, the same no-history setting. The differences are that it concerns online retail rather than a financial product, and that its inputs are real-time click-stream measures that STADIOEquities will only have if the SS1 data is supplied.

Public Dataset Used for Part B

Name	Bank Marketing dataset (bank-additional-full.csv)
Harvard reference	Moro, S., Rita, P. and Cortez, P. (2014) *Bank Marketing* [Dataset]. UCI Machine Learning Repository. doi: 10.24432/C5K306.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	41,188 instances in the bank-additional-full.csv file, supplied as a single file with no predefined training and test split. The older bank-full.csv version contains 45,211 instances and fewer features. The version used for this project is bank-additional-full.csv.
Number of feature columns	20 feature columns (plus the target column, y).
Number of classes	2 (yes and no). About 11.3% of instances are 'yes'.
Target range	Not applicable because this is a classification problem.
Origin	Real data from direct marketing campaigns, carried out by telephone, run by a Portuguese banking institution. The dataset is publicly available through the UCI Machine Learning Repository.

Notes on Dataset Choice and Modelling

The Bank Marketing dataset does not measure investment-account activation directly, but its binary target records whether a contacted customer converts to a financial product, so it is a suitable public proxy for testing the classification approach proposed for STADIOEquities.

Why it fits a limited-history problem. Of the 41,188 contacts, 35,563 (86.3%) had no earlier contact (*previous* = 0, *poutcome* = nonexistent) and 96.3% were recorded as not previously contacted (*pdays* = 999). Most of the rows are therefore first-time customers about whom the bank knew little, which is the situation for newly registered STADIOEquities customers. In this group the conversion rate is 8.8%, against 11.3% overall.

What it cannot supply. The dataset has no behaviour recorded after registration, such as sessions, logins or timing, which Sifa et al. (2015) and Sakar et al. (2019) both show to be valuable. These are the fields requested in the SS1 data request. For that reason the results on this dataset are used to prove the modelling approach, and are not expected to match the performance achievable on the client's data.

Alternative considered. The Online Shoppers Purchasing Intention dataset (Sakar and Kastro, 2018) has a one-session-per-user design and early-behaviour features, but it describes online retail rather than a financial product, so it was not selected.

Leakage control. The *duration* variable will be excluded during modelling because it represents information obtained during the marketing interaction and would not be available at the intended prediction point. Excluding it reduces the risk of target leakage and makes the approach more consistent with an early-prediction scenario.

References for the notes

Sakar, C. and Kastro, Y. (2018) *Online Shoppers Purchasing Intention Dataset* [Dataset]. UCI Machine Learning Repository. doi: 10.24432/C5F88Q.

Sakar, C.O., Polat, S.O., Katircioglu, M. and Kastro, Y. (2019) 'Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks', *Neural Computing and Applications*, 31, pp. 6893–6908. doi: 10.1007/s00521-018-3523-0.

Sifa, R., Hadiji, F., Runge, J., Drachen, A., Kersting, K. and Bauckhage, C. (2015) 'Predicting purchase decisions in mobile free-to-play games', *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, 11(1), pp. 79–85. doi: 10.1609/aiide.v11i1.12788.